

USER GUIDE · ENGLISH EDITION

# AFM Journal Search Tool v2

Integrated tool for AFM Manual Introduction writing:  
journal search · quality scoring · PDF collection · ZIP export

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Version v2.0 — Final Release

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2025

English User Guide

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# Tool Overview

## 1.1 What is AFM Journal Search Tool?

An integrated research tool designed to systematize **Chapter 1 Introduction writing** for AFM measurement mode manuals. It consolidates journal searching, evaluation, and organization into a **single HTML file**.

**Problems this tool solves**

Inconsistent search criteria, no unified IF·citations·equation evaluation, and repeated keyword-search-notes-draft cycles — all solved in one place.

## 1.2 Key Features

**Single HTML File**

No install. Opens instantly in Chrome/Edge. Most features work offline.

**API Auto-Search**

CrossRef + Unpaywall APIs for automatic keyword-based search and real-time PDF availability check.

**Auto Quality Scoring**

0–100 score calculated from IF, citations, equation count, and physics/circuit content.

**Batch ZIP Export**

Metadata, report, draft template, checklist, and PDFs — all in one ZIP.

## 1.3 System Requirements

| Item     | Minimum                    | Recommended                    |
|----------|----------------------------|--------------------------------|
| Browser  | Chrome 90+ / Edge 90+      | Latest Chrome                  |
| Internet | Required for auto-search   | For auto-search & PDF download |
| Email    | Required for Unpaywall API | Your active email              |
| OS       | Windows / macOS / Linux    | —                              |

# Getting Started

## 2.1 How to Open the Tool

### 1 Save the File

Save `AFM_Journal_Search_EN.html` to your computer.

### 2 Open in Browser

Drag and drop the HTML file into Chrome or Edge, or right-click → Open with → Chrome.

### 3 Ready to Use

All features available immediately — no login or installation required.

#### Important

Always open in **Chrome or Edge** for full feature support.

## 2.2 Interface Layout

| Section            | Menu Items                                             | Purpose            |
|--------------------|--------------------------------------------------------|--------------------|
| Workflow           | Keywords / Search Filters / Search Links               | Search preparation |
| Journal Management | Auto Registration / Manual Registration / Journal List | Journal collection |
| Finish             | ZIP Export / Write Process                             | Output & reference |

## Workflow — Step-by-Step Guide

### STEP 1 | Keyword Setup

Click **Keywords** in the sidebar. Select from 30 preset keywords or add custom ones.

#### Default active keywords

atomic force microscopy · tapping mode AFM · tip-sample interaction · cantilever dynamics · Q factor cantilever

### STEP 2 | Search Filter Configuration

Set publication year, IF, citation count, and result count. **Unpaywall API email is required** to enable auto registration.

### STEP 3 | Academic Database Search

| Database             | Specialization            | Sci-Hub |
|----------------------|---------------------------|---------|
| Google Scholar       | Broadest search           | —       |
| ScienceDirect        | Elsevier journals         | Yes     |
| Springer / Nature    | High-IF Nature family     | Yes     |
| ACS Publications     | American Chemical Society | Yes     |
| Wiley Online Library | Wiley journals            | Yes     |
| PubMed               | Bio/nano field            | —       |

### STEP 4 | Auto Journal Registration

#### 1 Click 'Start Auto Search'

CrossRef API searches top 10 → Unpaywall API verifies PDF availability for each

#### 2 Review Result Cards

Each card shows PDF Available or No PDF (Paywalled)

3

## Choose Registration Method

Register All with PDF / Register All / Individual registration

### Auto-filled fields

Title · Author · Year · Journal · Citations · DOI · PDF URL · OA status · Physics/Circuit detection · Quality Score

## STEP 5 | Manual Registration via DOI

Enter a DOI to auto-fill metadata and PDF URL from CrossRef and Unpaywall. Equation count must be manually counted from the paper's Introduction.

## STEP 6 | Journal List Management

### Sort Options

Score · Citations · IF · Year

### Selection Control

Checkbox determines ZIP inclusion

### PDF Quick Access

Open PDF directly from card button

### Statistics

Total · Selected · Has PDF · Avg Score

## STEP 7 | ZIP Export

1

### Enter AFM Mode Name

e.g. Tapping\_Mode, PFM, KPFM

2

### Select Files and Download

If PDFs included, each downloads sequentially before compression. May take several minutes.

# Journal Quality Score System

Citations40 pts

Max: 500+ citations —  $\min(\text{citations}/500, 1) \times 40$

Impact Factor30 pts

Max: IF 20+ —  $\min(\text{IF}/20, 1) \times 30$

Equations in Introduction20 pts

Max: 20+ equations —  $\min(\text{eq}/20, 1) \times 20$

Physics / Circuit included10 pts

+5 each — auto-detected from title & abstract keywords

| Score  | Grade         | Use                                |
|--------|---------------|------------------------------------|
| 70–100 | Excellent     | Primary reference for Introduction |
| 40–69  | Good          | Supplementary reference            |
| 0–39   | Review Needed | Verify before use                  |

## ZIP Output Files Detail

| Filename            |      | Format   | Contents                                                 |
|---------------------|------|----------|----------------------------------------------------------|
| journals.json       | JSON | JSON     | Full journal metadata (IF · citations · score · PDF URL) |
| report.html         | HTML | HTML     | Browser-viewable quality comparison report               |
| keywords.txt        | TXT  | TXT      | Keyword list and search query string                     |
| search_links.txt    | TXT  | TXT      | Auto-generated search URLs for 6 databases               |
| intro_template.md   | MD   | Markdown | Chapter 1 Introduction draft (with key equations)        |
| review_checklist.md | MD   | Markdown | AFM expert review checklist                              |
| pdf/ folder         | PDF  | PDF      | Auto-downloaded open-access PDFs                         |

## API Reference for Auto Registration

### CrossRef API

**Purpose:** Keyword search, DOI metadata

**Cost:** Free

**API Key:** Not required

### Unpaywall API

**Purpose:** Open-access PDF URL check

**Cost:** Free

**Required:** Email address only

### Email Privacy

Your email is sent only to the Unpaywall API server and not stored in the tool. Unpaywall does not use it for marketing.

## Frequently Asked Questions

### Q1. Can I use it without internet?

Keyword setup, manual registration, list management, and ZIP (without PDFs) work offline. Auto-search, DOI fetch, and PDF download require internet.

### Q2. Auto search returns no results.

Check: email entered, keywords selected (blue), internet connected, try broader keywords.

### Q3. Some journals show No PDF.

Journal is paywalled. Use Sci-Hub button with DOI or access through institutional account.

### Q4. ZIP is very slow.

Uncheck "pdf/ folder" for much faster ZIP generation without PDFs.

### Q5. Data disappears when browser closes.

Yes — current version uses session memory. Always export to ZIP before closing.

## Next Version Roadmap

### REVISION 1

#### Advanced Journal Grading Rules

- Differentiate equation weight by type (basic vs. core physics equations)
- Add field relevance index (auto-detect AFM topic alignment)
- Citation context analysis — core vs. incidental citations
- Tiered Impact Factor scoring with finer grade intervals
- Recency index — incorporate recent 5-year citation ratio

### REVISION 2

#### AI-Assisted Introduction Drafting from Registered Journals

- Auto-extract key sentences and summaries from registered journals
- Map content to the existing Introduction section structure
- Auto-map equations and references, generate draft
- User review → feedback → regeneration workflow
- Export final draft as Markdown or Word document

#### Contact & Feedback

Contact your Park Systems representative with issues or suggestions. User feedback is directly incorporated into the next version.